SUBJECT GUIDE

Computer Science — master course

One course for BPSC TRE 4.0, DSSSB TGT, KVS/NVS, UGC NET and technical posts: study each topic once, then add your exam's general sections.

12	77	32	65
Modules	Topics	NCERT chapters	Practice MCQs

Asked in: BPSC TRE · DSSSB TGT/PGT · KVS · NVS · UGC NET · SSC IMD · NIC

Contents

1 Computer fundamentals & digital logic
2 Programming & data structures
3 Operating systems
4 DBMS & SQL
5 Computer networks & security
6 Computer organisation & architecture
7 Theory of computation & compiler design
8 Algorithms
9 Software engineering & OOP (C++)
10 Web technology & internet
11 Discrete mathematics, statistics & numerical methods
12 Teaching & general sections

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MODULE 1

Computer fundamentals & digital logic

- ☐ Number systems: binary, octal, decimal, hexadecimal
- ☐ Boolean algebra, logic gates, truth tables
- ☐ Combinational & sequential circuits
- ☐ Binary arithmetic, 1's/2's complement, codes (BCD, Gray, ASCII, Unicode)
- ☐ K-map and Boolean simplification
- ☐ Flip-flops (SR, JK, D, T), registers, counters, multiplexers

🔥 ASKED AGAIN AND AGAIN

Base conversions & 2's complement | De Morgan's theorem | NAND/NOR as universal gates | JK flip-flop (J = K = 1 toggles) | Counters (mod-n) & multiplexer select lines

How it's asked: Asked as quick calculations: convert numbers, find a 2's complement, give a gate's output. NAND/NOR are universal gates.

READ (FREE, OFFICIAL)

- NCERT Class 11 · Computer System · PDF ↗
- NCERT Class 11 · Encoding Schemes and Number System · PDF ↗
- NCERT Class 11 · Computer System · PDF ↗
- NIOS NIOS Computer Science (330) ↗
- OFFICIAL NPTEL – free IIT/IISc courses ↗

▼ ✍️ Quick check — 7 MCQs (tap an option)

Q1. $(11010)_2$ in decimal is:

A 22 | B 28 | C 26 | D 24

Q2. Which pair are universal gates?

A XOR & XNOR | B AND & OR | C NAND & NOR | D NOT & AND

Q3. By De Morgan's theorem, (A·B)′ equals:

A A′·B′ | B A′ + B′ | C (A + B)′ | D A + B

Q4. A JK flip-flop with J = 1 and K = 1 at the clock edge will:

A Reset to 0 | B Toggle its output | C Hold its state | D Set to 1

Q5. How many select lines does an 8 : 1 multiplexer need?

A 3 | B 4 | C 8 | D 2

Q6. The 4-bit 2's complement of 0101 is:

A 1011 | B 0110 | C 1010 | D 1101

Q7. **A 3-bit binary counter has how many distinct states?**

A 6

B 3

C 16

D 8

Answers: 1-C 2-C 3-B 4-B 5-A 6-A 7-D

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MODULE 2

Programming & data structures

- ☐ Algorithms, flowcharts, pseudocode
- ☐ C: operators, loops, functions, pointers, recursion, arrays, structures, file handling
- ☐ Stack, queue, linked list, hashing
- ☐ C++/Java OOP basics
- ☐ Python: data types, control flow, functions, strings, lists, tuples, dictionaries
- ☐ Exception & file handling
- ☐ Trees, BST, heaps, graphs; BFS & DFS
- ☐ Output prediction & debugging MCQs

🔥 ASKED AGAIN AND AGAIN

Python output questions (//, %, slicing, len) · Stack/queue applications · Infix → postfix · BST worst case O(n) · Arrays vs linked lists

How it's asked: The biggest scorer. Expect "What is the output?" snippets — run small programs yourself instead of only reading.

READ (FREE, OFFICIAL)

- NCERT Class 11 · Introduction to Problem Solving · PDF ↗
- NCERT Class 11 · Getting Started with Python · PDF ↗
- NCERT Class 11 · Flow of Control · PDF ↗
- NCERT Class 11 · Functions · PDF ↗
- NCERT Class 11 · Strings · PDF ↗
- NCERT Class 11 · Lists · PDF ↗
- NCERT Class 11 · Tuples and Dictionaries · PDF ↗
- NCERT Class 12 · Exception Handling in Python · PDF ↗
- NCERT Class 12 · File Handling in Python · PDF ↗
- NCERT Class 12 · Stack · PDF ↗
- NCERT Class 12 · Queue · PDF ↗
- OFFICIAL Official Python tutorial ↗
- NIOS NIOS Computer Science (330) ↗

▼ ✍️ Quick check — 6 MCQs (tap an option)

Q1. Worst-case search time in a binary search tree with n nodes is:

A O(n log n) · B O(n) · C O(log n) · D O(1)

Q2. The postfix form of A + B * C is:

A ABC*+ · B AB+C* · C A+BC* · D ABC+*

Q3. In Python, print(7 // 2) outputs:

A 1 · B 3 · C 3.5 · D 4

Q4. In Python, len([1, [2, 3], 4]) is:

A Error · B 4 · C 2 · D 3

Q5. **Which data structure follows FIFO?**

A Tree

B Queue

C Graph

D Stack

Q6. **Accessing the i-th element in O(1) time is possible in:**

A Binary search tree

B Array

C Singly linked list

D Queue built on a linked list

Answers: 1-B 2-A 3-B 4-D 5-B 6-B

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MODULE 3

Operating systems

- ☐ Processes & threads
- ☐ Synchronisation, semaphores, critical section
- ☐ Paging, segmentation, virtual memory
- ☐ Thrashing; internal vs external fragmentation, compaction
- ☐ Linux basics & commands
- ☐ CPU scheduling (FCFS, SJF, Round Robin, priority)
- ☐ Deadlocks: conditions, prevention, banker's algorithm
- ☐ Page replacement (FIFO, LRU, Optimal)
- ☐ File systems & disk scheduling

🔥 ASKED AGAIN AND AGAIN

Four deadlock conditions & Banker's algorithm | FIFO vs Optimal vs LRU (Belady's anomaly) | SJF / Round Robin waiting time | Thrashing | Internal vs external fragmentation

How it's asked: Numericals are common: average waiting time for a schedule, number of page faults for FIFO/LRU.

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▼ ✍️ Quick check — 7 MCQs (tap an option)

Q1. **Which is NOT one of the necessary conditions for deadlock?**

A Hold and wait | B Circular wait | C Mutual exclusion | D Preemption

Q2. **Banker's algorithm is used for deadlock:**

A Recovery | B Detection | C Avoidance | D Prevention

Q3. **Belady's anomaly can occur with which page-replacement algorithm?**

A Optimal | B FIFO | C None of these | D LRU

Q4. **Thrashing happens when:**

A The cache is full | B The system spends more time paging than executing | C The CPU is idle waiting for I/O | D Two processes deadlock

Q5. **For jobs arriving together, which non-preemptive policy gives the minimum average waiting time?**

A Round Robin

B SJF

C FCFS

D Priority

Q6. **External fragmentation can be removed by:**

A Swapping the CPU

B Thrashing

C Increasing time quantum

D Compaction

Q7. **Round Robin with a very large time quantum behaves like:**

A Multilevel queue

B SJF

C Priority

D FCFS

Answers: 1-D 2-C 3-B 4-B 5-B 6-D 7-D

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MODULE 4

DBMS & SQL

- ☐ ER diagrams & relational model
- ☐ SQL: SELECT, inner & outer joins, GROUP BY/HAVING, subqueries, aggregate functions
- ☐ Transactions, ACID, concurrency control, isolation levels (read uncommitted → serializable)
- ☐ Keys, constraints, relational algebra
- ☐ Functional dependencies, Armstrong's axioms & normalisation (1NF–BCNF, 3NF vs BCNF)
- ☐ Indexing, B-trees and B+ trees

🔥 ASKED AGAIN AND AGAIN

Functional dependencies & Armstrong's axioms | 3NF vs BCNF | Outer joins (LEFT/RIGHT/FULL) | Isolation levels & anomalies | ACID | Indexing, B+ trees

How it's asked: Learn the clause order SELECT … FROM … WHERE … GROUP BY … HAVING … ORDER BY; normal-form and key questions repeat every year.

READ (FREE, OFFICIAL)

- NCERT Class 12 · Understanding Data · PDF ↗
- NCERT Class 12 · Database Concepts · PDF ↗
- NCERT Class 12 · Structured Query Language (SQL) · PDF ↗
- NCERT Class 11 · Database Concepts · PDF ↗
- NCERT Class 11 · Introduction to Structured Query Language (SQL) · PDF ↗
- NCERT Class 12 · Querying and SQL Functions · PDF ↗
- OFFICIAL NPTEL – free IIT/IISc courses ↗

▼ ✍️ Quick check — 7 MCQs (tap an option)

Q1. **Which statement is true?**

A BCNF allows partial dependencies

B BCNF and 3NF are identical

C Every 3NF relation is in BCNF

D Every BCNF relation is in 3NF

Q2. **A LEFT OUTER JOIN returns:**

A All rows of the left table, with NULLs where the right has no match

B Only matching rows

C The Cartesian product

D All rows of the right table only

Q3. **The lowest isolation level that prevents dirty reads is:**

A Repeatable Read

B Read Committed

C Read Uncommitted

D Serializable

Q4. **Which ACID property guarantees that committed data survives a crash?**

A Isolation

B Consistency

C Durability

D Atomicity

Q5. **Which clause filters groups after GROUP BY?**

A ORDER BY
B WHERE
C DISTINCT
D HAVING

Q6. **If A → B and B → C, then A → C. This Armstrong axiom is:**

A Union
B Augmentation
C Reflexivity
D Transitivity

Q7. **Most DBMS indexes for range queries are built as:**

A B+ trees
B Hash tables
C Linked lists
D Stacks

Answers: 1-D 2-A 3-B 4-C 5-D 6-D 7-A

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MODULE 5

Computer networks & security

- ☐ OSI and TCP/IP models
- ☐ IPv4/IPv6, IP classes, subnetting, routing
- ☐ ARP/RARP, ICMP; switching, sliding-window flow control, error & congestion control
- ☐ Cryptography, authentication, digital signatures, firewalls
- ☐ Ethernet, LAN/MAN/WAN, network devices
- ☐ TCP, UDP, DNS, HTTP, FTP, SMTP, DHCP and port numbers
- ☐ Medium access: CSMA/CD (Ethernet) vs CSMA/CA (Wi-Fi)

🔥 ASKED AGAIN AND AGAIN

OSI layer ↔ protocol ↔ device · ARP / RARP · IPv4 subnetting (hosts per subnet) · Sliding window flow control · CSMA/CD (Ethernet) vs CSMA/CA (Wi-Fi) · Port numbers

How it's asked: Match the layer ↔ protocol ↔ device; memorise common port numbers (HTTP 80, HTTPS 443, FTP 21, SMTP 25, DNS 53).

READ (FREE, OFFICIAL)

- NCERT Class 12 · Computer Networks · PDF ↗
- NCERT Class 12 · Data Communication · PDF ↗
- NCERT Class 12 · Security Aspects · PDF ↗
- NCERT Class 12 · Internet and Web · PDF ↗
- OFFICIAL NPTEL – free IIT/IISc courses ↗

▼ ✍️ Quick check — 7 MCQs (tap an option)

Q1. **ARP is used to find:**

A The port from a service name
B The MAC address from an IP address
C The domain name from an IP address
D The IP address from a MAC address

Q2. **How many usable host addresses does a /26 IPv4 subnet have?**

A 62
B 64
C 30
D 126

Q3. **The sliding-window protocol is mainly used for:**

A Routing
B Flow control
C Name resolution
D Encryption

Q4. **CSMA/CD is the access method of:**

A Bluetooth
B Token Ring
C Wi-Fi (802.11)
D Classic Ethernet (802.3)

Q5. **Routing is a function of which OSI layer?**

A Session
B Network
C Transport
D Data link

Q6. **The default port of HTTPS is:**

A 21
B 443
C 80
D 25

Q7. **Which device forwards frames using MAC addresses?**

A Repeater
B Hub
C Switch
D Router

Answers: 1-B 2-A 3-B 4-D 5-B 6-B 7-C

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MODULE 6

Computer organisation & architecture

- ☐ CPU, ALU, control unit
- ☐ Registers, buses, instruction cycle
- ☐ Cache, RAM, ROM, memory hierarchy
- ☐ Instruction formats & addressing modes
- ☐ Pipelining & hazards
- ☐ Interrupts, DMA, I/O organisation

🔥 ASKED AGAIN AND AGAIN

Addressing modes | Pipeline hazards | Memory hierarchy & cache | DMA & interrupts | Registers (PC, IR, MAR, MDR)

How it's asked: Expect addressing-mode identification and cache numericals (hit ratio, average access time).

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NCERT Class 11 · Computer System · PDF ↗

NIOS NIOS Computer Science (330) ↗

OFFICIAL NPTEL – free IIT/IISc courses ↗

▼ ✍️ Quick check — 5 MCQs (tap an option)

Q1. **In which addressing mode is the operand itself part of the instruction?**

A Register
B Direct
C Indirect
D Immediate

Q2. **A hazard caused by a branch instruction in a pipeline is a:**

A Control hazard
B Data hazard
C Structural hazard
D Memory hazard

Q3. **The Program Counter holds:**

A The stack top
B The current instruction
C The address of the next instruction
D The result of the ALU

Q4. **DMA allows:**

A The CPU to run faster
B Faster cache hits
C Direct access to registers
D I/O devices to transfer data to memory without the CPU moving each word

Q5. **The fastest storage in a computer is:**

A Main memory
B Registers
C SSD
D Cache

Answers: 1-D 2-A 3-C 4-D 5-B

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MODULE 7

Theory of computation & compiler design

- ☐ Sets, relations, functions, formal languages
- ☐ Regular expressions & finite automata (DFA, NFA)
- ☐ Context-free grammars & pushdown automata
- ☐ Turing machines & decidability
- ☐ Phases of a compiler; lexical analysis, parsing, syntax trees
- ☐ Intermediate code & runtime environments

🔥 ASKED AGAIN AND AGAIN

Phases of a compiler | DFA vs NFA (equal power) | Regular → finite automata, CFL → PDA | Lexical vs syntax analysis

How it's asked: Mostly for UGC NET / PGT-level papers: minimum DFA states and "which language is regular?" questions.

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▼ ✍️ **Quick check — 5 MCQs** (tap an option)

Q1. **Which compiler phase produces tokens?**

A Syntax analysis
B Code generation
C Code optimisation
D Lexical analysis

Q2. **Finite automata accept exactly the:**

A Recursively enumerable languages
B Regular languages
C Context-free languages
D Context-sensitive languages

Q3. **Compared with DFAs, NFAs are:**

A More powerful
B Less powerful
C Equally powerful
D Unable to accept regular languages

Q4. **A parse tree is built during:**

A Linking
B Loading
C Syntax analysis
D Lexical analysis

Q5. **A pushdown automaton recognises:**

A All languages

B Regular languages only

C Only finite languages

D Context-free languages

Answers: 1-D 2-B 3-C 4-C 5-D

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MODULE 8

Algorithms

- ☐ Time & space complexity; Big-O, Ω, Θ
- ☐ Sorting algorithms and their complexity
- ☐ Greedy algorithms & dynamic programming
- ☐ Linear & binary search
- ☐ Recursion, divide and conquer
- ☐ BFS, DFS, shortest paths (Dijkstra vs Bellman-Ford), minimum spanning trees (Kruskal, Prim)

🔥 ASKED AGAIN AND AGAIN

Quick sort vs merge sort complexity | Dijkstra vs Bellman-Ford | BFS vs DFS | Binary search (sorted, O(log n)) | Greedy (Kruskal, Prim) vs DP

How it's asked: Keep one table of best/average/worst complexity for every sort and search — it answers many questions.

READ (FREE, OFFICIAL)

NCERT Class 12 · Sorting · PDF ↗

NCERT Class 12 · Searching · PDF ↗

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▼ ✍️ Quick check — 6 MCQs (tap an option)

Q1. Dijkstra's algorithm can give wrong results when the graph has:

A Negative edge weights
B More than 100 vertices
C Cycles
D Undirected edges

Q2. Worst-case time complexity of quick sort is:

A O(log n)
B O(n)
C $O(n^2)$
D O(n log n)

Q3. Merge sort runs in:

A $O(n^2)$ in the worst case
B O(n) on sorted input
C O(log n)
D O(n log n) in all cases

Q4. Binary search requires the data to be:

A Stored in a linked list
B Sorted
C Of size a power of two
D Unique

Q5. **Kruskal's minimum spanning tree algorithm is a:**

A Dynamic-programming algorithm

B Greedy algorithm

C Backtracking algorithm

D Divide-and-conquer algorithm

Q6. **Shortest paths (fewest edges) in an unweighted graph are found with:**

A BFS

B Heap sort

C Binary search

D DFS

Answers: 1-A 2-C 3-D 4-B 5-B 6-A

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MODULE 9

Software engineering & OOP (C++)

- ☐ Encapsulation, inheritance, polymorphism, abstraction
- ☐ C++: overloading vs overriding, virtual functions, abstract classes, virtual destructors, constructors
- ☐ UML & object-oriented design
- ☐ SDLC, waterfall, spiral, Agile
- ☐ Requirements, testing (black/white box), debugging, maintenance
- ☐ Metrics: cyclomatic complexity, cohesion & coupling, COCOMO

🔥 ASKED AGAIN AND AGAIN

Cyclomatic complexity (E − N + 2P) | High cohesion, low coupling | Spiral model = risk analysis | Overloading vs overriding | Virtual functions, abstract classes, virtual destructors

How it's asked: Definitions and "which model suits…" questions; know each testing type and SDLC model by its key feature.

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▼ ✍️ Quick check — 7 MCQs (tap an option)

Q1. Cyclomatic complexity of a control-flow graph with E edges, N nodes and P components is:

A N − E + 2P | B E − N + P | C E + N − 2P | D E − N + 2P

Q2. A good software design has:

A Low coupling, low cohesion | B High coupling, low cohesion | C Low coupling, high cohesion | D High coupling, high cohesion

Q3. The distinguishing feature of the spiral model is:

A Explicit risk analysis in every cycle | B Single pass through phases | C No testing phase | D No customer feedback

Q4. In C++, runtime polymorphism is achieved using:

A Templates | B Virtual functions | C Macros | D Function overloading

Q5. **A C++ class with at least one pure virtual function is:**

A A template class

B A final class

C A friend class

D An abstract class

Q6. **A base class destructor should be virtual so that:**

A Memory is never freed

B Deleting a derived object through a base pointer also runs the derived destructor

C Objects are created faster

D Constructors become virtual

Q7. **Function overriding means:**

A Same name, different parameters in one class

B A derived class redefines a base-class function with the same signature

C Two classes with the same name

D Calling a function recursively

Answers: 1-D 2-C 3-A 4-B 5-D 6-B 7-B

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MODULE 10

Web technology & internet

- ☐ HTML, CSS, JavaScript, DOM
- ☐ XML, DTD, parsers
- ☐ Website design, publishing, basic web security
- ☐ Client–server architecture, web servers
- ☐ HTTP, browsers, internet services
- ☐ E-commerce & mobile computing

🔥 ASKED AGAIN AND AGAIN

HTTP status codes | HTML tags & attributes | CSS selectors & properties | DOM, client vs server side

How it's asked: Tag/attribute and HTTP status-code questions (200, 301, 404, 500) are easy marks.

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NCERT Class 12 · Internet and Web · PDF ↗

OFFICIAL MDN – Learn web development ↗

OFFICIAL W3C web standards ↗

▼ ✍️ Quick check — 4 MCQs (tap an option)

Q1. **HTTP status code 404 means:**

A OK

B Moved permanently

C Not found

D Internal server error

Q2. **DOM stands for:**

A Digital Ordinance Model

B Document Object Model

C Document Oriented Markup

D Data Object Model

Q3. **Which HTML tag creates a hyperlink?**

A <href>

B <nav>

C <link>

D <a>

Q4. **The CSS property that sets text colour is:**

A foreground

B font-color

C text-color

D color

Answers: 1-C 2-B 3-D 4-D

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MODULE 11

Discrete mathematics, statistics & numerical methods

- ☐ Logic, sets, relations, functions, counting
- ☐ Graph theory & trees
- ☐ Matrices, numerical errors, interpolation
- ☐ Permutations, combinations, probability
- ☐ Descriptive statistics & distributions
- ☐ Numerical methods & basic calculus

🔥 ASKED AGAIN AND AGAIN

Permutations with repetition | Probability with dice/cards | Trees have n − 1 edges | Truth tables of implication

How it's asked: Probability and counting questions are short — practise formulas until they are automatic.

READ (FREE, OFFICIAL)

NCERT Class 10 · Statistics · PDF ↗

NCERT Class 10 · Probability · PDF ↗

NCERT Class 11 · Measures of Central Tendency · PDF ↗

NCERT Class 11 · Correlation · PDF ↗

NIOS NIOS Mathematics (311) ↗

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▼ ✍️ Quick check — 4 MCQs (tap an option)

Q1. **Number of distinct arrangements of the letters of BOOK is:**

A 24 | B 8 | C 12 | D 6

Q2. **Probability of a sum of 7 when two dice are thrown is:**

A 1/9 | B 7/36 | C 1/6 | D 1/12

Q3. **A tree with n vertices has how many edges?**

A 2n | B n − 1 | C n + 1 | D n

Q4. **The implication p → q is false only when:**

A p false, q true | B p true, q false | C p true, q true | D p false, q false

Answers: 1-C 2-C 3-B 4-B

12

MODULE 12

Teaching & general sections

- ☐ Teaching aptitude & pedagogy; child development where prescribed
- ☐ Hindi and English
- ☐ ICT & educational technology
- ☐ Classroom management & assessment
- ☐ General awareness, reasoning, quantitative aptitude

🔥 ASKED AGAIN AND AGAIN

Language section is qualifying — secure the cut-off | General Studies counts for merit

How it's asked: These sections decide selection in close contests — give them 15 minutes every day.

READ (FREE, OFFICIAL)

ON EDQITY History, Geography & other subject guides →

ON EDQITY Current affairs capsule →

ON EDQITY NCERT books →

How to use this for each exam

EXAM	MAIN FOCUS
BPSC TRE 4.0 Computer Science	Technical CS + prescribed language and general studies sections
DSSSB TGT Computer Science	Technical syllabus + the prescribed general section
KVS / NVS teaching	Technical CS + the relevant teaching and general sections
UGC NET Computer Science	Deeper theory, algorithms, mathematics and research aptitude — check eligibility
NIC / technical recruitments	Core CS: algorithms, OS, DBMS, networks + the post's own syllabus

Daily routine · 3 hours

90 min

Learn one technical topic

Understand concepts, formulas, diagrams and examples.

45 min

Solve MCQs & previous-year questions

Practise questions from teaching and technical exams.

30 min

Revise and record mistakes

Keep short notes, formulas and a wrong-answer notebook.

15 min

Teaching / general section

Rotate through the extra sections your target exams require.

Practise

BPSC TRE 4.0 CS mock tests →

DSSSB TGT CS mock tests →

SSC IMD Computer Science mocks →

BPSC TRE 4.0 CS exam kit →

Official syllabus

BPSC – official syllabus page ↗

DSSSB TGT Computer Science – Directorate of Education (PDF) ↗

DSSSB TGT Computer Science – NDMC (PDF) ↗

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